# YI LI

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## RESEARCH AND ENGINEERING FOCUS

Pretraining data for frontier LLMs at trillion-token scale: web-scale processing, quality filtering, interleaved web corpora, token efficiency, and training-time robustness.

## WORK EXPERIENCE

**xAI, Pretraining** June 2025 – May 2026
*Member of Technical Staff*

- Owned pretraining web data pipelines; processed trillions of web tokens for Grok pretraining.
- Built a web-interleaved data pipeline producing ∼1T tokens for pretraining and image generation.
- Designed and shipped text and image quality filters to improve data quality and token efficiency.
- Instrumented metrics to track abnormal sequences during training and applied online masking to improve training stability.
- Curated hundreds of billions of high-quality tokens from X, Wikipedia, and Wikimedia.

**NVIDIA, AI Robotics Research Lab** Jan. 2024 – June 2025
*Research Intern* *supervised by Dr. Ankit Goyal*

- Developed HAMSTER, a hierarchical Vision-Language-Action (VLA) model for open-world robot manipulation that enabled visual, semantic, and geometric generalization across domain gaps. Accepted to ICLR 2025.

**Microsoft Research Asia, Visual Computing Group** Nov. 2015 – Jun. 2017
*Research Intern* *supervised by Dr. Jifeng Dai*

- Developed Deformable Convolutional Networks, introducing deformable convolution and RoI pooling for geometric variation in images. Accepted to ICCV 2017 (oral).
- Developed FCIS, an end-to-end instance segmentation framework accepted to CVPR 2017 (spotlight); won 1st Prize in the MS COCO 2016 Segmentation Challenge.
- Developed R-FCN, a fast and accurate region-based fully convolutional object detector accepted to NeurIPS 2016.

## EDUCATION

**University of Washington** *2018 – 2025*
Ph.D., advised by Prof. Dieter Fox — Computer Science and Engineering
**Tsinghua University** *2014 – 2017*
Master of Science, with highest distinction — Automation
**Tsinghua University** *2010 – 2014*
Bachelor of Engineering (National Scholarship) — Automation

## SELECTED PUBLICATIONS

(* indicates co-first author, full list at Google Scholar)

Yi Li*, Yuquan Deng*, Jesse Zhang*, Joel Jang, Marius Memmel, Caelan Garrett, Fabio Ramos, Dieter Fox, Anqi Li, Abhishek Gupta, Ankit Goyal
**HAMSTER: Hierarchical Action Models for Open-World Robot Manipulation**
*International Conference on Learning Representations (ICLR)*, 2025. Project website
**Invited presentations at OpenAI, Google DeepMind, and ByteDance Seed**

Yi Li, Gu Wang, Xiangyang Ji, Yu Xiang, Dieter Fox
**DeepIM: Deep Iterative Matching for Object Pose Estimation**
In *International Journal of Computer Vision (IJCV)*, 2020, arXiv
In *European Conference on Computer Vision (ECCV)*, 2018 (oral)
**Selected as one of the top 12 papers in ECCV 2018**

Jifeng Dai*, Haozhi Qi*, Yuwen Xiong*, Yi Li*, Guodong Zhang*, Han Hu, Yichen Wei
**Deformable Convolutional Networks**
In *International Conference on Computer Vision (ICCV)*, 2017 (oral). arXiv
**Ranked 6th among the most influential papers in ICCV 2017**

Yi Li*, Haozhi Qi*, Jifeng Dai, Xiangyang Ji, Yichen Wei
**Fully Convolutional Instance-aware Semantic Segmentation**
In *Computer Vision and Pattern Recognition (CVPR)*, 2017 (spotlight). arXiv
1st Prize in MS COCO 2016 Segmentation Challenge

Jifeng Dai, Yi Li, Kaiming He, Jian Sun
**R-FCN: Object Detection via Region-Based Fully Convolutional Networks**
In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016. arXiv
**Ranked 2nd among the most influential papers in NeurIPS 2016**

## PROFESSIONAL SERVICE AND HONORS

Served as a reviewer for top-tier conferences and journals, including ICLR, CVPR, ICCV, ECCV, RA-L, AAAI, ICRA, and IROS, since 2017.

Outstanding 2017 Master's Thesis, awarded by the Chinese Institute of Electronics (top 10 in China)

1st Prize in MS COCO 2016 Object Detection Challenge

Outstanding 2016 Intern at MSRA

2017 Summa Cum Laude in Beijing (top 2% at Tsinghua)

2013 National Scholarship (top 0.5% nationwide)